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## Lab 3: Vending machine

### Goals

- Design sequential systems using Verilog
- Design state machines using Verilog

### Introducción

A finite state machine (FSM) allows modeling systems in which the outputs depend on the current state and the inputs. FSMs are fundamental for the design of digital circuits. In this laboratory, we will design a vending machine that accepts coins and dispenses a product when the correct amount has been inserted.

### Lab description

Design a state machine that models the behavior of a vending machine. The vending machine will have the following characteristics:

- The vending machine will offer 4 products at different prices, as shown in Table I.
- To start a purchase, the **init** button must be pressed.
- Once the **init** button is pressed, the vending machine will accept coins; for this implementation, **it will only accept \$1 coins**, and it will not accept more than \$9.
- The amount entered must be displayed on a 7-segment display.
- Once the amount for the desired product is entered, **the product is selected**, and then the **dispense button** is pressed to dispense the product.
- If the amount of money entered is less than the cost of the product, the machine will not dispense the product; it will wait until either a lower-priced product is selected and the dispense button is pressed again or additional coins are inserted.
- If the amount of money is greater than the cost of the selected product and the dispense button is pressed, **the motor will be activated** to push the product out of the machine; **the motor will deactivate once the product falls**.
- If money has been inserted and the purchase is to be canceled, simply press the **cancel button**; in that case, the money will be returned.
- Once the purchase is completed or canceled, the machine will **wait for the change and purchased product to be collected**, and it will display the change on a 7-segment display.
- Once the products and change have been collected, the process will end, and the machine will return to its initial state.

| Producto | Costo |
|----------|-------|
| 00       | 4     |
| 01       | 9     |
| 10       | 2     |
| 11       | 7     |

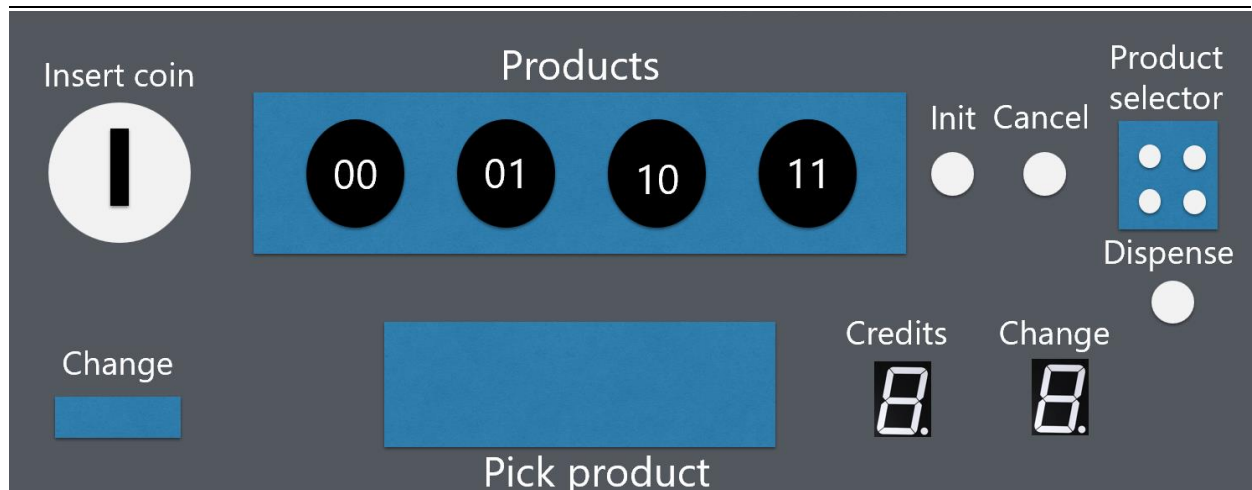


Figura 1. Sketch of the proposed vending machine.

The finite state machine that will model the vending machine will have the following states:

**IDLE:** In this state, the machine does not perform any action.

**INSERT\_COIN:** In this state, the machine accepts coins.

**DISPENSE:** This state indicates that a product has been purchased.

**PICK\_PRODUCT:** This state indicates that the product and change have been dispensed.

The state diagram in Figure 2 describes the operation of the vending machine.

To simulate the elements of the vending machine, use the following FPGA components:

**Start Purchase:** To start the purchase, use a button.

**Insert Coin:** To insert a coin, use a button.

**Select Product:** To select the product, use 2 switches.

**Dispense:** To dispense the product, use a button.

**Cancel Purchase:** To cancel the purchase, use a button.

**Product Dispensed:** To simulate that the product has been dispensed, use a switch (This signal in a real project can be obtained using a sensor).

**Collect Product/Change:** To simulate the collection of the product and change, use a switch (This signal in a real project can be obtained using a sensor).

**Motors:** To simulate the activation of the vending machine's motors, use LEDs.

**States:** In order to observe the operation of the state machine, it is suggested to add outputs that indicate the current state using LEDs.

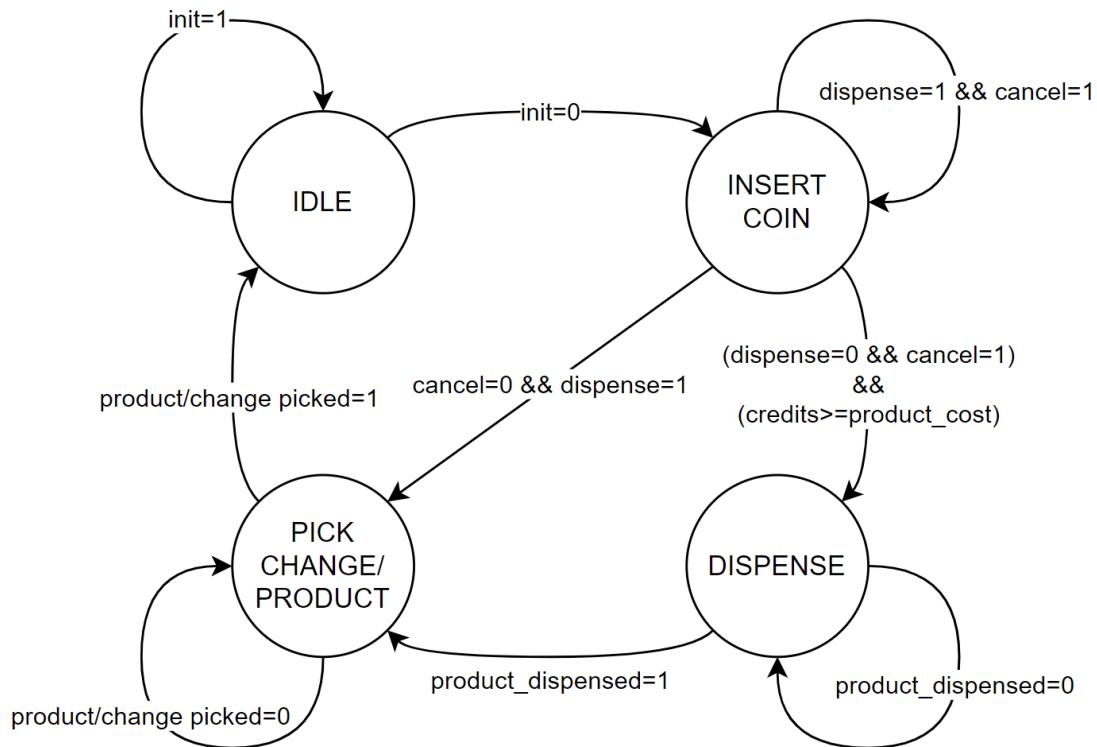


Figura 2. Vending machine state diagram.

### Extra logic

The state machine is the main circuit of our vending machine; however, it is not the only one. During each state, the following actions must be performed:

**IDLE:** Se deben reiniciar todos los displays de 7 segmentos a 0.

**INSERT\_COIN:** A count of the inserted coins must be maintained; for this, the use of a counter is suggested (this circuit will be provided by the instructor). Additionally, the inserted coins should be displayed on a 7-segment display.

**DISPENSE:** The motor that pushes the product out of the machine must be activated.

**PICK\_CHANGE/PRODUCT:** The change corresponding to the completed purchase must be calculated and displayed on a 7-segment display, based on the previous state.

## Submission

### Implementation description

Generate a document providing a detailed description of the implementation. This should include an explanation of design decisions, challenges encountered, and how they were overcome. Include diagrams, schematics, tables, or equations used in the design development.

**Video:**

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Upload a short video to the Teams group, recording the operation of the implemented circuit on the FPGA and providing a brief description of its functionality.

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